

Cyberbullying Detection in Smart Cities: Project Proposal

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Abstract

Cyberbullying is increasingly embedded in digital public life and can undermine social well-being in smart-city ecosystems where citizens rely on online platforms for daily services and communication. This project proposes a reproducible text-based cyberbullying detection pipeline designed for smart-city contexts, where scale, topic drift, and class imbalance are common. We will benchmark a unified set of *baseline models* that includes both classical machine learning classifiers and modern transformer encoders fine-tuned on cyberbullying data. Building on these baselines, we propose *our model*: a context-enriched transformer that incorporates domain-adaptive pretraining and imbalance-aware learning to improve robustness and generalization. Beyond basic accuracy-oriented reporting, we will evaluate using macro/weighted F1, AUROC/AUPRC, calibration, statistical significance tests, and targeted error analyses. Deliverables include a final PDF proposal, an implementation repository, and an experimental report documenting model performance, efficiency, and reliability under smart-city-like deployment constraints.

Cyberbullying Detection in Smart Cities: Project Proposal

Introduction and Motivation

Smart-city services depend on online interaction at scale (e.g., community groups, public feedback platforms, and service apps). In these spaces, cyberbullying and harassment can spread quickly and harm individuals as well as public trust. Automated detection is therefore important for early risk screening and for supporting moderation workflows.

In practice, cyberbullying detection is challenging for at least three reasons. First, user-generated content is informal and often includes sarcasm, code-mixing, and rapidly changing slang. Second, labels are usually highly imbalanced: “non-bullying” dominates, while certain bullying types appear only rarely. Third, reported results may not generalize because datasets differ in platform style, annotation policy, and time period. Recent survey work highlights that performance often drops under cross-dataset testing and that evaluation practices are not always comparable [[1]]. A dedicated analysis of limitations also points to reproducibility gaps and data scarcity as recurring issues [[2]]. Motivated by these observations, our proposal emphasizes (i) a clean benchmark, (ii) a clear “our model” improvement target, and (iii) richer evaluation that better matches deployment needs.

Problem Statement

Given a short social-media style post, our goal is to classify whether it contains cyberbullying content. Depending on the dataset, the task can be:

- **Binary:** bullying vs. non-bullying; or
- **Multi-class:** bullying categories such as gender, ethnicity, religion, age, or other targeted harassment classes.

Our proposal focuses on a text-first approach, while keeping the design extensible to multi-modal signals (images, user context, or interaction graph) in future iterations.

Objectives and Research Questions

Objectives

1. Build a reproducible preprocessing and training pipeline for cyberbullying detection.
2. Establish strong baselines that include both classic ML and transformer fine-tuning under the same protocol.
3. Propose and test an improved model that targets robustness under class imbalance and domain shift.
4. Report results using richer evaluation methods (beyond accuracy/precision/recall/F1 alone), including calibration and statistical testing.

Research Questions

1. How do classic ML baselines compare with transformer baselines when data splits, tuning budget, and reporting metrics are controlled?
2. Does domain adaptation plus imbalance-aware training improve macro-level performance and reduce failure cases on minority bullying types?
3. Which evaluation perspectives (AUPRC, calibration error, robustness checks) provide more actionable insight than single-point metrics?

Related Work

Early cyberbullying detection systems relied on lexical features and traditional classifiers such as Logistic Regression and SVM. More recent approaches use deep neural architectures (e.g., recurrent networks) and contextual language models, where transformers have become a default starting point in many text classification tasks. BERT [[3]] and RoBERTa [[4]] can capture context and long-range dependencies and often outperform bag-of-words models, especially when bullying cues depend on phrase context.

However, surveys emphasize that “better models” do not automatically translate to reliable systems [[1]]. Domain mismatch (platform style, region, slang) can reduce performance, and label imbalance can lead to inflated accuracy while minority classes are poorly detected. Domain-adaptive pretraining (continued pretraining on in-domain text before task fine-tuning) is one practical strategy for closing domain gaps [[5]]. Meanwhile, imbalance-aware learning methods such as focal loss re-weight difficult and under-represented examples and are widely used when positive classes are rare [[6]]. Our proposal combines these ideas into one coherent experimental plan: treat classic ML and transformer fine-tuning as baselines, then test a single “our model” that focuses on robustness.

Proposed Methodology

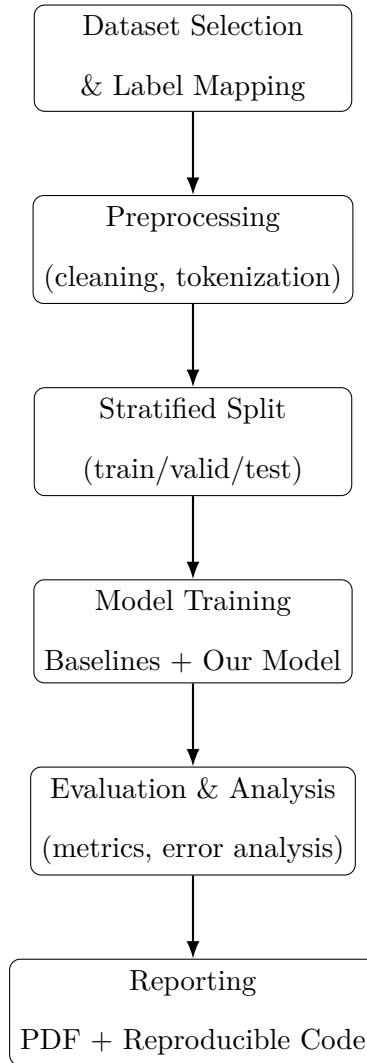
Figure 1 summarizes the overall workflow.

Dataset and Data Preparation

We will start from a public cyberbullying dataset consisting of short posts and labels indicating bullying content and/or bullying type. A common structure is one majority “non-bullying” class and multiple minority harassment categories. This matches our smart-city motivation because citizen-facing platforms often contain a large amount of normal content with a smaller but important subset of harmful interactions.

Data preparation will follow a consistent and documented protocol:

- **Data audit:** check label distribution, duplicates, and obvious noise; define a stable label mapping for the final task.
- **Normalization:** normalize URLs and user mentions, handle excessive repeated characters, and standardize emojis when possible.
- **Tokenization:** TF-IDF features for classic ML; subword tokenization for transformer baselines and our model.

**Figure 1**

End-to-end workflow for the proposed cyberbullying detection project.

- **Splitting:** stratified train/validation/test split; if multiple datasets exist, reserve one for out-of-domain evaluation to approximate domain shift.

To reduce accidental leakage, we will de-duplicate near-identical texts before splitting and keep a clear random seed policy.

Baseline Models

1. **Classic ML baselines:** Logistic Regression and Linear SVM trained on TF-IDF features (word and character n-grams).

2. **Transformer baselines:** Fine-tuned BERT [[3]] and RoBERTa [[4]] with a standard classification head.

Baselines will share the same data split and be tuned using a fixed hyperparameter budget, to keep comparisons fair.

Our Model (Proposed Improvement)

Our model builds on the transformer baseline but targets two practical weaknesses: domain mismatch and minority-class errors. We propose the following enhancements:

- a) **Domain-adaptive pretraining (DAPT):** continue masked-language-model pretraining on an in-domain corpus (e.g., platform- or topic-specific text related to smart-city services) before fine-tuning [[5]]. The aim is to adapt vocabulary and style without changing the downstream task definition.
- b) **Imbalance-aware learning:** apply focal-style re-weighting or class-balanced weights to emphasize hard/minority examples [[6]]. We will report how this affects minority-class recall and macro-F1.
- c) **Robust training recipe:** early stopping on macro-F1, learning-rate scheduling, and simple data augmentation (e.g., random deletion or punctuation noise) to reduce sensitivity to minor text variation.

Training Protocol and Implementation Details

We will implement the pipeline in Python with versioned dependencies and configuration files. Key protocol decisions include:

- **Hyperparameter tuning:** grid/random search under a fixed budget (same budget for all transformer models).
- **Reproducibility:** fixed random seeds, logged experiments, and saved checkpoints.

Table 1*Model families included in the proposal.*

Category	Model	Key design
Baseline	LR / Linear SVM	TF-IDF features; fast training; interpretable weights; strong linear benchmark.
Baseline	BERT / RoBERTa	Fine-tuning with classification head; contextual representations and transfer learning.
Our Model	DAPT + Imbalance-aware Transformer	Continued in-domain pretraining + loss re-weighting to improve minority-class and domain robustness.

- **Efficiency reporting:** training time per epoch and inference speed on a standard hardware setup (to reflect deployment constraints).

These design choices are motivated by concerns raised in reproducibility-focused analyses of cyberbullying research [[2]].

Evaluation Plan

1. **Imbalance-aware metrics:** macro-F1, weighted-F1, per-class precision/recall, and confusion matrices (focus on minority classes).
2. **Ranking/threshold metrics:** AUROC and AUPRC, with threshold tuning for a selected operating point (e.g., high recall for screening) [[1]].
3. **Calibration:** Expected Calibration Error (ECE) and reliability diagrams, to assess whether predicted probabilities are trustworthy.
4. **Robustness checks:** performance under noisy inputs (typos, emoji, character

flooding) and (if feasible) cross-dataset evaluation.

- 5. **Statistical reliability:** bootstrap confidence intervals and paired significance tests (e.g., McNemar-style tests) to support claims [[7]].
- 6. **Error analysis:** targeted qualitative analysis of false positives/negatives, with a focus on minority harassment types and ambiguous language.

This evaluation plan is intended to produce a more complete picture than single overall accuracy and to align better with real moderation needs.

Project Timeline and Work Allocation

The project follows a 15-week schedule starting from January 9th, 2026, aligned with the course requirements. The detailed phases and deliverables are outlined in Table 2 and visualized in Figure 2.

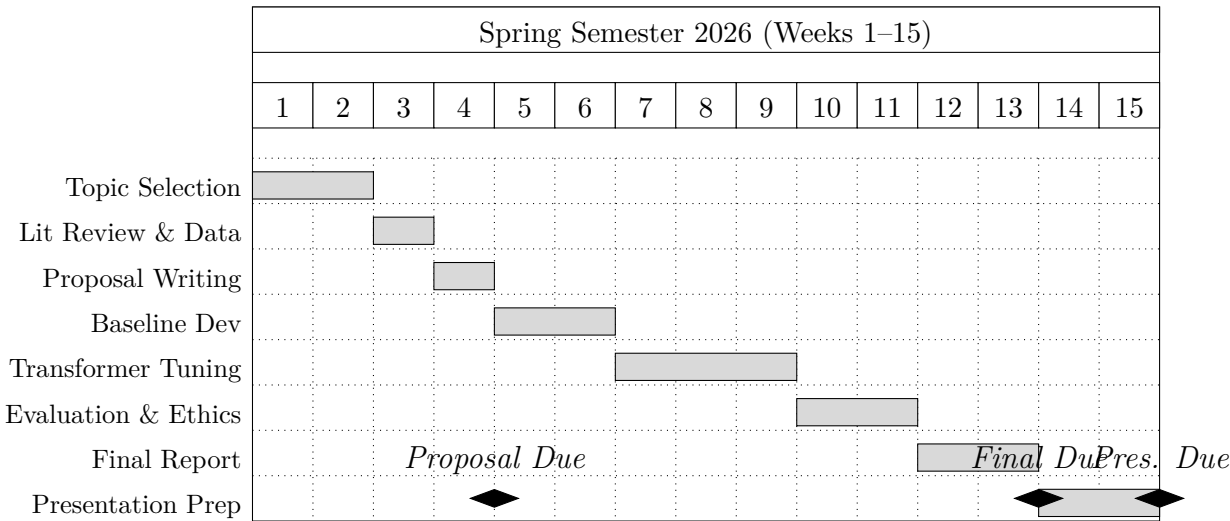


Figure 2
Gantt chart of the project lifecycle, from initialization to final presentation.

Expected Outcomes

By the end of the project, we expect to deliver:

Table 2*Project Schedule (Jan 9 – May 3, 2026)*

Week	Date Range	Key Tasks	Deliverables
1–2	Jan 9 – Jan 26	Team formation, topic confirmation, and scope definition.	Topic Confirmation Form, Work Division Plan.
3	Jan 27 – Feb 2	Deep literature review, dataset collection, and preprocessing (cleaning/labeling).	Lit Review Report, Preprocessed Dataset.
4	Feb 3 – Feb 9	Writing the Project Proposal (1500–2000 words).	Final Proposal (30% Grade).
5–6	Feb 10 – Feb 23	Feature engineering; Building and training Baseline models (LR/SVM).	Baseline Model Files, Training Log/Report.
7–9	Feb 24 – Mar 15	Fine-tuning Transformers (BERT/RoBERTa); Parameter optimization using GPU.	Transformer Models, Model Comparison Report.
10–11	Mar 18 – Mar 29	Multi-dimensional evaluation, error analysis, and ethics/compliance check.	Evaluation Report, Optimization Plan.
12–13	Apr 1 – Apr 12	Expanding Proposal into the Final Report; Turnitin plagiarism check.	Final Project Report (50% Grade).
14–15	Apr 15 – May 3	Preparing presentation (20–30 mins) and rehearsal; Final archiving.	Presentation PPT, Archived Project Assets (20% Grade).

1. A cleaned dataset split and a fully reproducible training pipeline (scripts + configuration).
2. A benchmark report comparing unified baselines (classic ML + transformer fine-tuning) under the same protocol.
3. Our improved model implementation and a documented ablation study showing the effect of domain adaptation and imbalance-aware learning.
4. A comprehensive evaluation package including AUPRC/AUROC, calibration analysis, robustness checks, and statistically supported comparisons.
5. A final proposal PDF suitable for grading, with clear figures and tables.

Risks and Mitigation

- **Label imbalance and unstable minority metrics:** we will report macro-F1 and confidence intervals, and we will tune thresholds based on AUPRC rather than accuracy alone.
- **Domain shift:** if an out-of-domain dataset is not available, we will simulate shift with topic-based splits or time-based splits where possible, and report the impact.
- **Reproducibility issues:** we will maintain fixed seeds, versioned dependencies, and experiment logs, following reproducibility recommendations [[2]].

Conclusion

This proposal targets a practical and reproducible cyberbullying detection pipeline for smart-city digital environments. By treating all comparison systems as baselines and emphasizing a single improved “our model,” the experimental structure becomes clearer and easier to grade. We also commit to richer evaluation methods that better reflect the realities of imbalanced and shifting online language. The resulting artifacts will support

both course assessment and future extensions toward multi-modal or platform-integrated smart-city moderation tools.

References

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